

May 11, 2026

1 Part 1

The objective of this part was to process a given corrupted audio signal . The signal contained two main components: a human voice (100–250 Hz) and a musical instrument (600–900 Hz). However, it was affected by a high-frequency jamming tone and environmental noise.

To identify the interference, a global Fourier Transform (FFT) was performed on the entire signal. By analyzing the magnitude spectrum, a very sharp and high-magnitude peak was observed at approximately **16000.00** Hz. Since this frequency is significantly higher than the 900 Hz limit of the musical instrument, it was classified as the jamming tone.

To remove the identified jamming tone, a sharp **Low-pass filter** was designed.

- **Type:** 4th order Butterworth filter (using Second-Order Sections for stability).
- **Cutoff Frequency:** 1000 Hz.

This filter successfully eliminated the high-frequency jammer while preserving all relevant information below 1000 Hz, including the voice and the instrument.

After cleaning the signal, two **Bandpass filters** were applied to separate the components:

1. **Human Voice:** Isolated between 100 Hz and 250 Hz.
2. **Musical Instrument:** Isolated between 600 Hz and 900 Hz.

The resulting magnitude spectra confirmed that each component was successfully extracted without interference from the other.

The Signal-to-Noise Ratio (SNR) was calculated to measure the success of the filtering process for the voice component.

- **SNR Before Filtering:** 0.00 dB
- **SNR After Filtering:** 39.12 dB
- **SNR Improvement:** 39.12 dB

The significant increase in SNR proves that the noise was reduced effectively while keeping the target voice signal.

The effectiveness of these filtering stages can be visually observed in Figure 1, which displays the magnitude spectra at each step. So, Fourier analysis allowed us to pinpoint the interference frequency accurately.

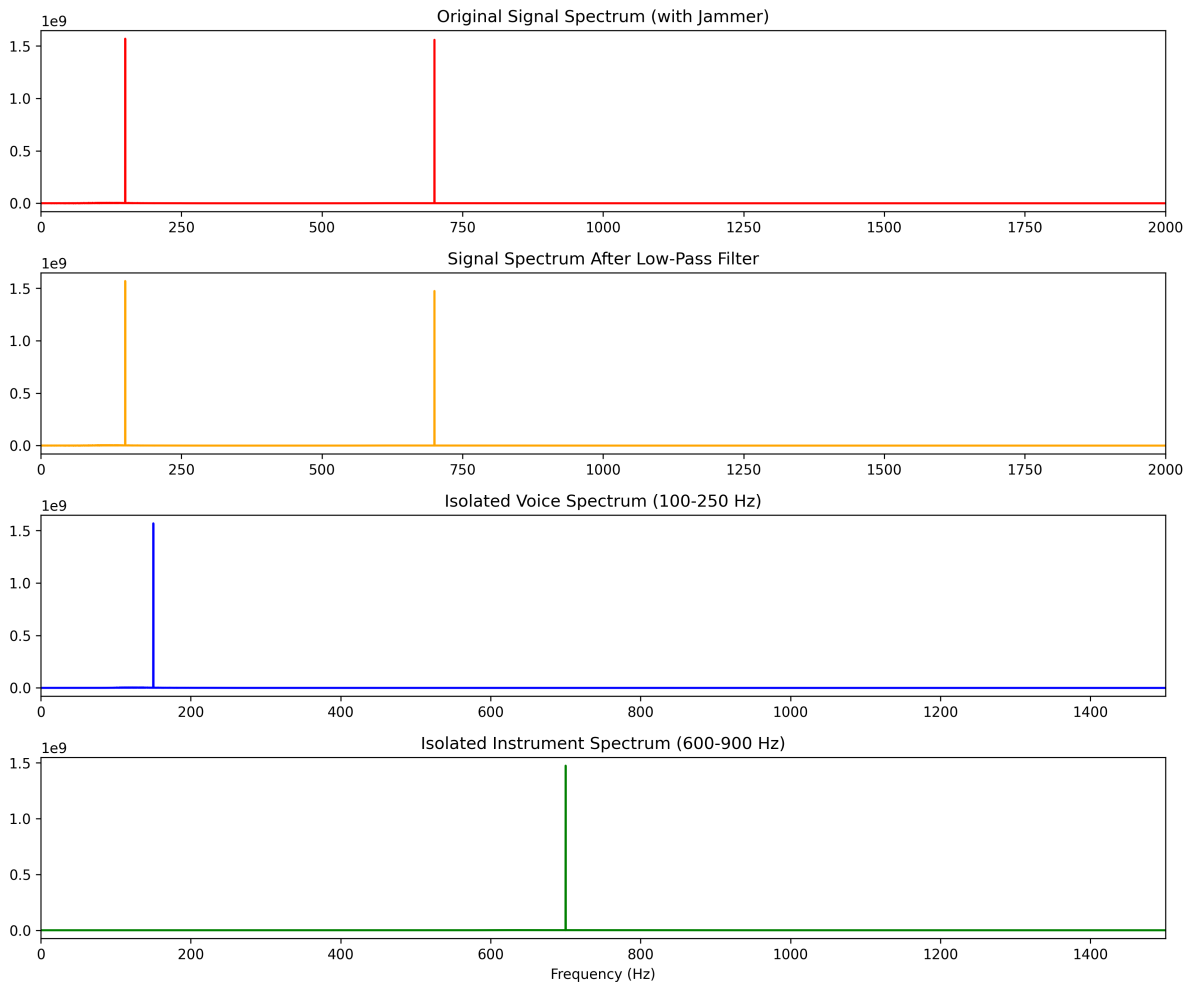


Figure 1: Magnitude spectra of original, filtered, and isolated signals.

2 Part 2

In this part there are three main processing steps:

1. **Loading Data:** the voice signal, the instrument signal, and the room impulse response is loaded .
2. **Convolution:** the room effect is applied to the voice using convolution . This adds the room's echoes to the clean voice.
3. **Reconstruction:** Because convolution makes the signal longer, zero-padding is added to the instrument track. Then i mixed the reverberant voice with the padded instrument to create the hybrid signal.

Fast Fourier Transform (FFT) is used to compare the magnitude spectra of the signals. The original voice had a smooth spectrum, but the processed voice showed many periodic "dips" or nulls. These dips are the characteristic sign of a room's acoustics.

The sharp dips in the spectrum are caused by **destructive interference**. In the time domain, the impulse response contains multiple echoes. When these delayed echoes are added to the original voice, they cancel out specific frequencies. This effect is known as **comb filtering**. The spacing of these dips depends on the timing of the echoes in the room impulse response.

By using linear convolution, a real acoustic environment is successfully simulated. The results show that a room acts as a complex filter that changes the frequency balance of a sound based on its physical reflections.

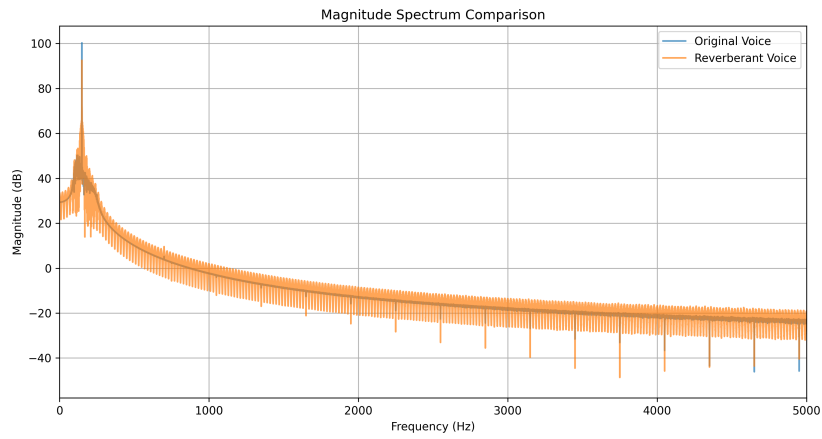


Figure 2: Magnitude Spectrum Comparison: Original vs. Reverberant Voice

3 Part 3

Global DFT

First, I computed the global Discrete Fourier Transform (DFT) for the entire 3-second mystery file. However, this visualization fails to show the order of the waveforms. The standard Fourier Transform removes all time information; it only tells us what frequencies exist in the total signal, not when they occur.

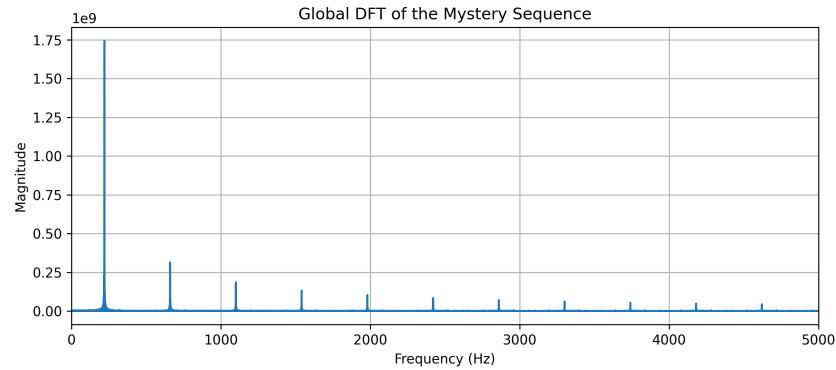


Figure 3: Global DFT of the entire 3-second mystery sequence.

STFT and Transition Timestamps

To observe the frequencies over time, I implemented a Short-Time Fourier Transform (STFT) to generate a spectrogram. Based on this spectrogram, I identified that the waveform changes exactly at $t = 1.0$ seconds and $t = 2.0$ seconds.

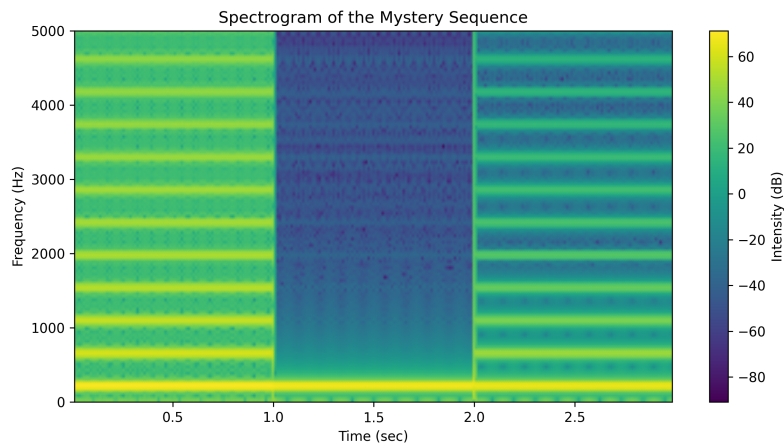


Figure 4: Spectrogram of the mystery sequence showing transitions at 1.0s and 2.0s.

Therefore, I divided the 3-second mystery sequence into three equal 1-second segments. By analyzing the DFT of each individual segment, I classified the waveforms as follows:

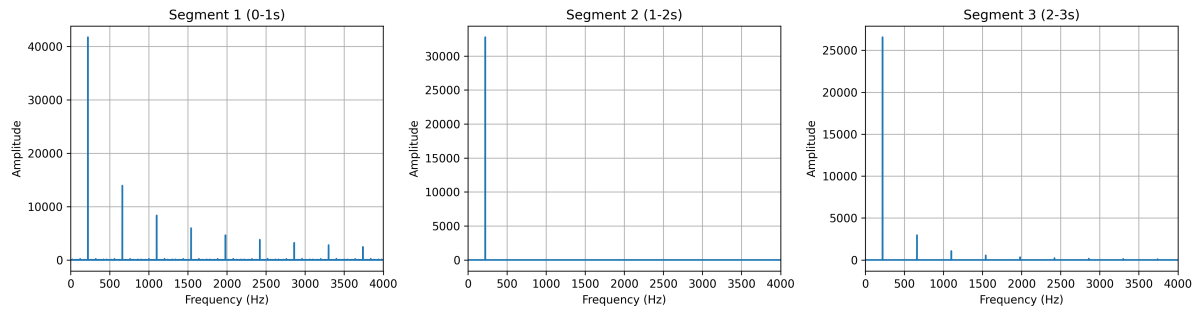


Figure 5: Frequency spectrum for each 1-second segment.

Segment 1 (0.0s - 1.0s): Square Wave

The frequency spectrum for the first segment shows a fundamental frequency along with multiple odd harmonics. The amplitude of these harmonics decreases relatively slowly. Mathematically, the amplitude decay of these harmonics follows a $1/n$ pattern (where n is the harmonic number). This specific slow decay rate is a defining characteristic of a **Square Wave**.

Segment 2 (1.0s - 2.0s): Sine Wave

The spectrum for the second segment is very simple. It displays only a single, strong peak at the fundamental frequency. There are no additional peaks or harmonics present in the higher frequencies. Because it contains only one frequency without any harmonic energy, I can confidently classify this segment as a **Sine Wave**.

Segment 3 (2.0s - 3.0s): Triangle Wave

Similar to the first segment, the third segment also contains a fundamental frequency and odd harmonics. However, the key difference is the harmonic decay rate. In this segment, the amplitude of the harmonics drops very rapidly. This fast decrease follows a $1/n^2$ pattern. For example, the third harmonic is only $1/9$ of the fundamental amplitude. This rapid decay rate proves that this segment is a **Triangle Wave**.